

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I Dillilokesh D 192311097 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dillilokesh D

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Answers

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Representation

Step 1 — Define the atomic propositions (symbols):

Symbol	Meaning (atomic proposition)
F	Patient has Fever
C	Patient has Cough
Rs	Patient has Rash
B	Patient has Breathlessness
Flu	Patient possibly has Flu
Ms	Patient possibly has Measles
Pn	Patient possibly has Pneumonia

Step 2 — Encode the rules using these symbols:

Rule	Propositional Logic Sentence
R1	$F \wedge C \rightarrow \text{Flu}$
R2	$F \wedge \text{Rs} \rightarrow \text{Ms}$
R3	$C \wedge B \rightarrow \text{Pn}$

Step 3 — Encode the facts about Patient A as literal assignments:

F = True C = True B = True Rs = (not stated $\Rightarrow$ assumed False / Unknown)

Note on completeness

Rash (Rs) is never mentioned for Patient A, so under the Closed-World Assumption used by production/expert systems, Rs is treated as False. This is stated explicitly because it changes whether R2 can fire (see part b).

(b) Forward Chaining — Deriving All Diagnoses for Patient A

Forward chaining is data-driven: at every iteration we scan all rules and fire any whose premises are already known True, adding the conclusion to the working memory. We repeat until a full pass produces no new facts (fixpoint).

Iter.	Rule Tested	Premise Evaluation	Fires?	Fact Derived
1	R1: $F \wedge C \rightarrow \text{Flu}$	$F=\text{True}, C=\text{True} \Rightarrow T \wedge T = \text{True}$	Yes	$\text{Flu} = \text{True}$
1	R2: $F \wedge R_s \rightarrow M_s$	$F=\text{True}, R_s=\text{False} \Rightarrow T \wedge F = \text{False}$	No	— (blocked)
1	R3: $C \wedge B \rightarrow P_n$	$C=\text{True}, B=\text{True} \Rightarrow T \wedge T = \text{True}$	Yes	$P_n = \text{True}$
2	R1, R2, R3 re-checked	No premise changed; Flu & Pn are not premises of any rule	No	No new facts — fixpoint reached

Result of Forward Chaining

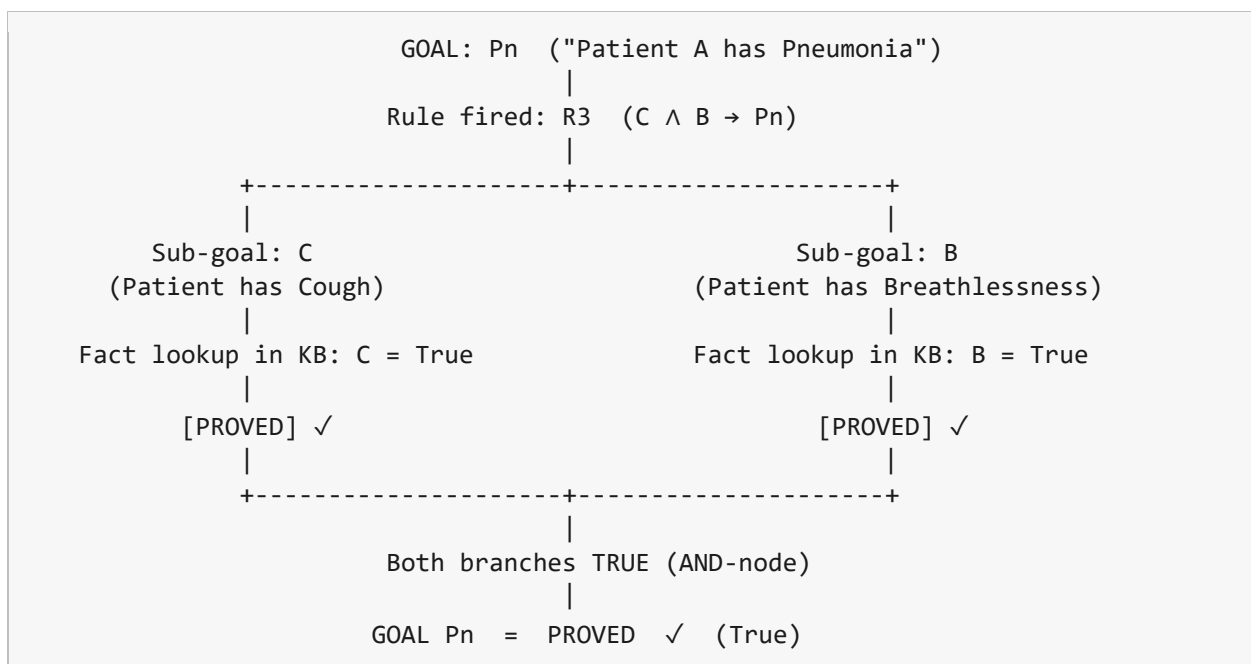
Working memory after termination = { F, C, B, Flu, Pn }. The agent concludes: Patient A possibly has Flu AND possibly has Pneumonia. Measles is ruled out because the Rash symptom is absent.

(c) Backward Chaining — Verifying the Goal 'Patient A has Pneumonia'

Backward chaining starts at the goal (Pn) and works backward, searching the KB for a rule whose conclusion matches the goal, then recursively proving each premise of that rule as a new sub-goal, until every branch bottoms out at a known fact.

Step	Current Goal	Rule Matched	Action
1	Pn (Pneumonia)	R3: $C \wedge B \rightarrow P_n$	Reduce goal Pn into two AND-sub-goals: C and B
2	Sub-goal: C	Fact lookup	$C = \text{True}$ in KB $\Rightarrow$ sub-goal PROVED directly
3	Sub-goal: B	Fact lookup	$B = \text{True}$ in KB $\Rightarrow$ sub-goal PROVED directly
4	Back to Pn	Both AND-branches proved	Pn is PROVED $\Rightarrow$ goal confirmed True

Goal-reduction tree:



Correctness check

This result is consistent with the forward-chaining result in part (b), where P_n was also derived as True. Forward and backward chaining are both sound and complete for definite-clause (Horn) knowledge bases such as this one, so agreement between the two methods confirms the derivation is correct.

Case Study 2 — Autonomous Traffic Management Agent

Because this KB reasons about relationships between different vehicles ($\text{Behind}(x, y)$) and quantifies over an unbounded domain of vehicles, First-Order Logic (FOL) is required — propositional logic cannot express 'for all vehicles x ' or a relation between two named objects.

Knowledge base recap:

Rule 1: $\forall x \text{ Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Rule 2: $\forall x \text{ ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

Rule 3: $\forall x \forall y \text{ Vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

Facts : $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

a) Unification of Rule 1 with the Facts

Rule 1's premise has two literals: $\text{Vehicle}(x)$ and $\text{EmergencyType}(x)$. We unify each against the matching fact, one substitution at a time, and confirm the substitutions are consistent (this consistency check is what makes the result a valid unifier).

Step	Unify	Substitution Applied	Resulting θ
1	$\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$	$x \rightarrow \text{Ambulance}$	$\theta_1 = \{x / \text{Ambulance}\}$
2	$\text{EmergencyType}(x)$ with $\text{EmergencyType}(\text{Ambulance})$	apply θ_1 : $\text{EmergencyType}(\text{Ambulance})$ already matches $\text{EmergencyType}(\text{Ambulance})$	$\theta_2 = \{x / \text{Ambulance}\}$ (consistent with θ_1)
3	Combine	θ_1 and θ_2 agree on $x \rightarrow \text{Ambulance}$, no conflict	MGU $\theta = \{x / \text{Ambulance}\}$

Applying θ to Rule 1 yields the fully instantiated (ground) implication:

$\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$

Both premises are known facts, so by Modus Ponens:

Derived fact

$\text{ClearPath}(\text{Ambulance}) = \text{True}$ [MGU used: $\theta = \{x / \text{Ambulance}\}$]

(b) Resolution Proof that $\text{Proceed}(\text{CarA})$ is True

Step 1 — Convert every rule to Conjunctive Normal Form (CNF)

Sentence	Eliminate $\rightarrow$ ($a \rightarrow b \equiv \neg a \vee b$)	Final CNF Clause
Rule 1	$\neg(\text{Vehicle}(x) \wedge \text{EmergencyType}(x)) \vee \text{ClearPath}(x)$	$\neg\text{Vehicle}(x) \vee \neg\text{EmergencyType}(x) \vee \text{ClearPath}(x)$
Rule 2	$\neg\text{ClearPath}(x) \vee (\text{GreenSignal}(x) \wedge \neg\text{RedSignal}(x))$	split into two clauses: $(\neg\text{ClearPath}(x) \vee \text{GreenSignal}(x))$ and $(\neg\text{ClearPath}(x) \vee \neg\text{RedSignal}(x))$
Rule 3	$\neg(\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y)) \vee \text{Proceed}(x)$	$\neg\text{Vehicle}(x) \vee \neg\text{Behind}(x,y) \vee \neg\text{GreenSignal}(y) \vee \text{Proceed}(x)$

The four facts are already atomic, so each is its own unit clause. The goal is negated for refutation:

Facts: $\text{Vehicle}(\text{Ambulance})$ $\text{EmergencyType}(\text{Ambulance})$ $\text{Vehicle}(\text{CarA})$ $\text{Behind}(\text{CarA}, \text{Ambulance})$

Negated goal: $\neg\text{Proceed}(\text{CarA})$

Step 2 — Resolution derivation

#	Resolve Clause	With Clause	θ (unifier)	Resolvent
1	$\neg\text{Vehicle}(x) \vee \neg\text{EmergencyType}(x) \vee \text{ClearPath}(x)$	$\text{Vehicle}(\text{Ambulance})$	$x/\text{Ambulance}$	$\neg\text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$
2	Result of (1)	$\text{EmergencyType}(\text{Ambulance})$	—	$\text{ClearPath}(\text{Ambulance})$
3	$\neg\text{ClearPath}(x) \vee \text{GreenSignal}(x)$	$\text{ClearPath}(\text{Ambulance})$ [from 2]	$x/\text{Ambulance}$	$\text{GreenSignal}(\text{Ambulance})$
4	$\neg\text{Vehicle}(x) \vee \neg\text{Behind}(x,y) \vee \neg\text{GreenSignal}(y) \vee \text{Proceed}(x)$	$\text{Vehicle}(\text{CarA})$	x/CarA	$\neg\text{Behind}(\text{CarA},y) \vee \neg\text{GreenSignal}(y) \vee \text{Proceed}(\text{CarA})$
5	Result of (4)	$\text{Behind}(\text{CarA}, \text{Ambulance})$	$y/\text{Ambulance}$	$\neg\text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$
6	Result of (5)	$\text{GreenSignal}(\text{Ambulance})$ [from 3]	—	$\text{Proceed}(\text{CarA})$
7	$\text{Proceed}(\text{CarA})$ [from 6]	$\neg\text{Proceed}(\text{CarA})$ (negated goal)	—	$\square$ (empty clause)

Conclusion of Resolution Proof

The derivation of the empty clause $\square$ in step 7 is a contradiction, which proves — by refutation — that $\text{KB} \models \text{Proceed}(\text{CarA})$. $\text{Proceed}(\text{CarA})$ is therefore logically entailed and TRUE: CarA is permitted to proceed because it is directly behind an ambulance that has been granted a green signal.

(c) Is the FOL Knowledge Base Sufficient for Two Simultaneous Emergency Vehicles?

Not entirely. The current KB is sufficient to grant a clear path and a green signal to any single emergency vehicle considered in isolation, and to let a following vehicle proceed behind it. It is not sufficient to safely resolve two emergency vehicles that are active at the same time, for the following logical reasons:

- No conflict / precedence predicate: Rule 1 fires independently for every x satisfying $\text{Vehicle}(x) \wedge \text{EmergencyType}(x)$. If two ambulances approach a shared intersection from crossing directions, Rule 1 and Rule 2 would derive $\text{GreenSignal}(x)$ for both simultaneously — an unsafe, contradictory real-world outcome that the logic does not flag as a conflict.
- No spatial/intersection relation: the KB has no predicate such as $\text{Intersects}(x, y)$ or $\text{AtJunction}(x, j)$ to detect that two vehicles' paths cross, so it cannot even recognise that a conflict exists.
- No priority ordering: real emergency dispatch uses priority (e.g., a fire truck responding to a structure fire may outrank an ambulance on a routine transfer). The KB has no $\text{Priority}(x, n)$ predicate or comparison rule.
- No temporal/state reasoning: FOL as used here is static — it has no notion of 'signal changes after t seconds' or 'clear path is released once vehicle has passed'. Two emergency vehicles arriving at different times need event/situation-calculus style time-indexed predicates, e.g. $\text{ClearPath}(x, t)$.

Recommended additions to the KB:

New Predicate / Rule	Purpose
$\text{Intersects}(x, y)$	Declares that vehicles x and y share a conflicting path segment.
$\text{Priority}(x, n)$	Assigns a numeric/ordinal priority level n to vehicle x (e.g., Fire Truck > Ambulance > Police).
$\forall x \forall y$ $\text{Vehicle}(x) \wedge \text{Vehicle}(y) \wedge \text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{Intersects}(x, y) \wedge \text{Priority}(x) > \text{Priority}(y) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$	Resolves conflicts by granting clearance only to the higher-priority vehicle while the lower-priority one waits.
$\text{ClearPath}(x, t), \text{GreenSignal}(x, t)$	Time-indexed versions of the existing predicates so that clearance can be granted, held, and released over time as vehicles arrive and pass.

Case Study 3 — Agricultural Expert Reasoning System

This is again a purely propositional KB (no objects/variables), so CNF conversion and resolution refutation can be carried out directly on the propositions given.

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
Facts: $\text{SoilDry} = \text{True}, \quad \text{CropWheat} = \text{True}$

(a) Conversion to Conjunctive Normal Form (CNF)

Each implication is converted using the standard three-step procedure: (1) eliminate $\rightarrow$ using $a \rightarrow b \equiv \neg a \vee b$, (2) push $\neg$ inward using De Morgan's laws, (3) distribute $\vee$ over $\wedge$ if needed. Since none of these sentences contain a conjunction inside a negation after step 2 except P3, only P3 needs the De Morgan step spelled out.

Sentence	Eliminate Implication	Push $\neg$ Inward (De Morgan)	Final CNF Clause
P1	$\text{SoilDry} \rightarrow \text{IrrigationNeeded} \equiv \neg \text{SoilDry} \vee \text{IrrigationNeeded}$	(no $\neg$ over $\wedge/\vee$ to push)	$(\neg \text{SoilDry} \vee \text{IrrigationNeeded})$
P2	$\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$	$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$	$(\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod})$
P3	$\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$	$(\text{ApplyDripMethod} \vee \neg \text{CropWheat}) \vee \text{CropAtRisk}$	$(\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk})$
Fact 1	SoilDry (already atomic)	—	(SoilDry)
Fact 2	CropWheat (already atomic)	—	(CropWheat)

CNF Clause Set

$C1 = (\neg \text{SoilDry} \vee \text{IrrigationNeeded})$ $C2 = (\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod})$ $C3 = (\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk})$ $C4 = (\text{SoilDry})$ $C5 = (\text{CropWheat})$

(b) Resolution Refutation — Proving 'ApplyDripMethod'

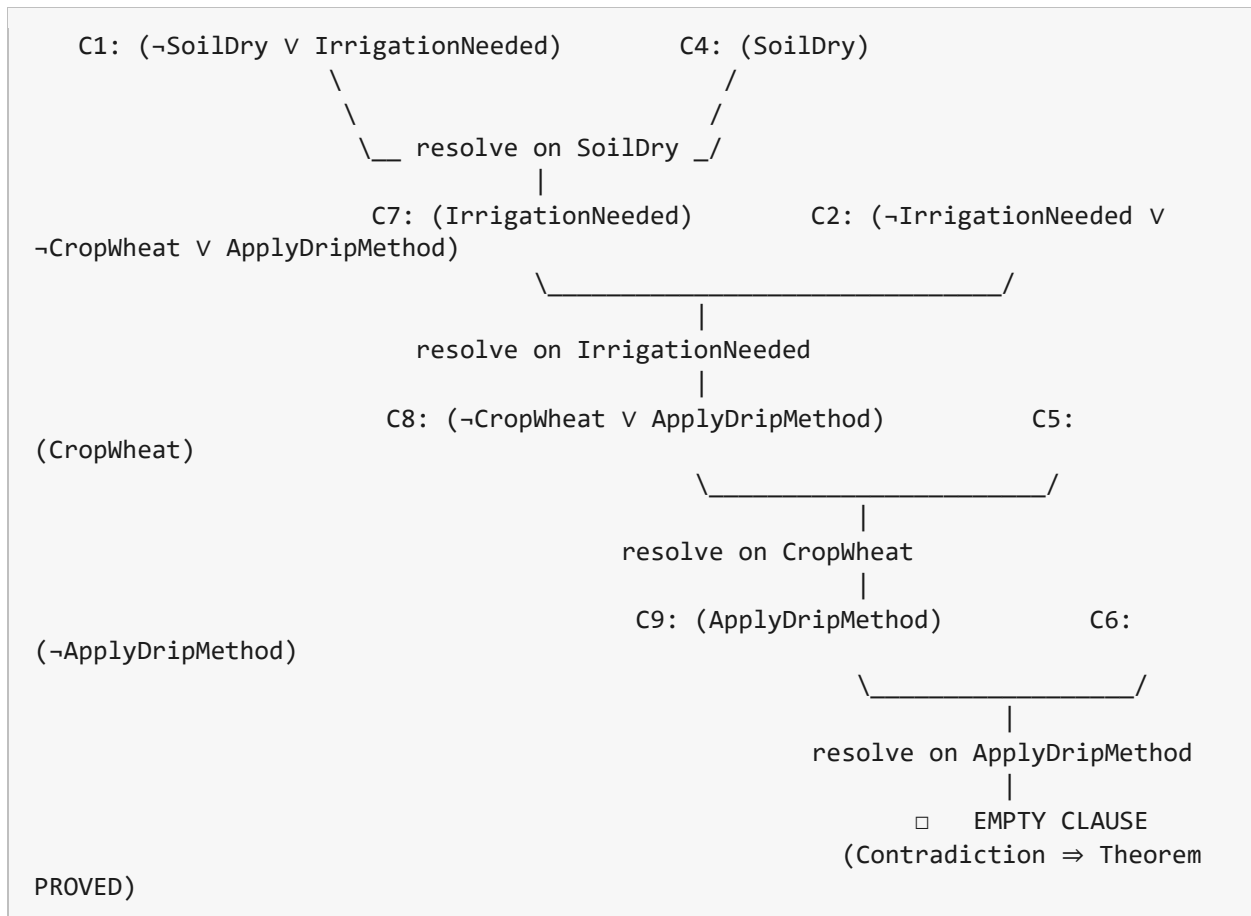
To prove a sentence α by refutation, we add $\neg \alpha$ to the CNF clause set and show that resolving clauses repeatedly yields the empty clause $\square$, i.e. $\text{KB} \wedge \neg \alpha$ is unsatisfiable, so $\text{KB} \models \alpha$.

Negated goal: $C6 = (\neg \text{ApplyDripMethod})$

Step	Resolve	With	Resolvent
1	$C1 = \neg \text{SoilDry} \vee \text{IrrigationNeeded}$	$C4 = \text{SoilDry}$	$C7 = \text{IrrigationNeeded}$

Step	Resolve	With	Resolvent
2	$C2 = \neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$	$C7 = \text{IrrigationNeeded}$	$C8 = \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
3	$C8 = \neg \text{CropWheat} \vee \text{ApplyDripMethod}$	$C5 = \text{CropWheat}$	$C9 = \text{ApplyDripMethod}$
4	$C9 = \text{ApplyDripMethod}$	$C6 = \neg \text{ApplyDripMethod}$	$\square$ (empty clause — contradiction)

Resolution proof tree:



Theorem proved

The empty clause $\square$ is derived, so $KB \wedge \neg \text{ApplyDripMethod}$ is unsatisfiable. Hence $KB \models \text{ApplyDripMethod}$: 'Apply Drip Method' is logically entailed and TRUE. This agrees with a direct forward-chaining check: $\text{SoilDry} \Rightarrow \text{IrrigationNeeded}$, and $\text{IrrigationNeeded} \wedge \text{CropWheat} \Rightarrow \text{ApplyDripMethod}$ — the same conclusion reached by a different, refutation-based route confirms correctness.

(c) Effect of Removing the Fact 'SoilDry'

If SoilDry is removed from the fact base, walk the same chain again:

Step	Reasoning	Result
1	P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$. SoilDry is no longer a known fact (neither True nor provably False).	IrrigationNeeded can NO LONGER be derived by Modus Ponens.

Step	Reasoning	Result
2	P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod. Its first conjunct (IrrigationNeeded) is now unproven.	ApplyDripMethod can NO LONGER be derived — the resolution chain from part (b) breaks at Step 1, since clause C4 (SoilDry) no longer exists to resolve with C1.
3	P3: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk. This rule needs $\neg$ ApplyDripMethod, not merely 'ApplyDripMethod unproven'.	Under strict monotonic (classical) logic, $\neg$ ApplyDripMethod is NOT entailed either — the sentence is simply undetermined, so CropAtRisk cannot be validly proved by classical resolution.
4	Under the Closed-World Assumption (CWA) / Negation-as-Failure — the convention typically used by rule-based expert systems such as this one — any proposition that cannot be derived as True is treated as False.	Under CWA: ApplyDripMethod is treated as False $\Rightarrow \neg$ ApplyDripMethod = True. Combined with CropWheat = True, rule P3's premise is satisfied $\Rightarrow$ CropAtRisk IS derived = True.

Correctness note — Monotonic vs Non-monotonic reasoning

This is a genuine and important distinction, not a minor technicality. Classical (monotonic) first-order/propositional resolution can never conclude CropAtRisk once SoilDry is removed, because $\neg$ ApplyDripMethod is not entailed by the remaining KB — it is merely absent. Most deployed expert systems, however, use negation-as-failure (a non-monotonic reasoning pattern, as in Prolog/production systems), under which 'not proved' is treated as 'false'. Under that convention CropAtRisk still follows. The agent's designer must decide — and document — which convention governs the system, because the two conventions give opposite answers to 'does CropAtRisk still follow?'

Case Study 4 — Wumpus World Knowledge Agent

The agent starts at [1,1] and moves along the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2], collecting percepts at each cell it visits. Grid neighbours are the orthogonally adjacent cells only (no diagonals), following the standard Wumpus World convention. Percepts collected along the path:

Cell Visited	Percepts Sensed There
[1,1]	Breeze = True, Stench = False, Glitter = False
[1,2]	Stench = True, Breeze = False, Glitter = False
[2,2]	Glitter = True, Stench = False, Breeze = False

(a) Belief State from Percepts — Modus Ponens

Modus Ponens: from $p \rightarrow q$ and p , infer q . Each rule is applied the instant its triggering percept is sensed.

#	Known Fact (p)	Rule ($p \rightarrow q$)	Modus Ponens Result (q)
1	Breeze[1,1] = True	Rule 2: Breeze[1,1] $\rightarrow$ Pit adjacent to [1,1]	Pit is in one (or more) of the neighbours of [1,1] = {[1,2], [2,1]}

#	Known Fact (p)	Rule (p → q)	Modus Ponens Result (q)
2	Stench[1,2] = True	Rule 1: Stench[1,2] → Wumpus adjacent to [1,2]	Wumpus is in one of the neighbours of [1,2] = {[1,1], [1,3], [2,2]}
3	Glitter[2,2] = True	Rule 3: Glitter[x,y] → Gold at [x,y]	Gold[2,2] = True (Gold is definitely at [2,2])
4	Agent occupies [1,1] unharmed	Occupied-cell convention: the cell the agent stands on contains neither Wumpus nor Pit	$\neg$ Wumpus[1,1] $\wedge$ $\neg$ Pit[1,1] $\Rightarrow$ by Rule 4, Safe[1,1] = True

All conclusions derived in (a)

Pit $\in$ {[1,2], [2,1]} | Wumpus $\in$ {[1,1], [1,3], [2,2]}, and since [1,1] is already confirmed occupied/safe, this narrows to Wumpus $\in$ {[1,3], [2,2]} | Gold[2,2] = True | Safe[1,1] = True

(b) Forward Chaining — Locating the Wumpus and the Pit

Forward chaining now proceeds cell-by-cell as the agent explores, combining each new percept with everything already known (including the important negative information — the absence of a stench or breeze also licenses an inference, via Rule 4 in its contrapositive form).

Step	New Percept / Position	Inference Fired	Fact Derived
1	At [1,1]: Breeze=T, Stench=F	Rule 2 (Breeze→Pit adjacent); Rule 1 contrapositive ($\neg$ Stench[1,1] $\Rightarrow$ no Wumpus adjacent to [1,1])	Pit $\in$ {[1,2],[2,1]}; Wumpus $\notin$ {[1,2],[2,1]}
2	Agent moves to [1,2] and survives	Occupied-cell convention	$\neg$ Pit[1,2] $\wedge$ $\neg$ Wumpus[1,2] confirmed $\Rightarrow$ Safe[1,2] = True
3	Combine Step 1 + Step 2	Elimination: Pit $\in$ {[1,2],[2,1]} and $\neg$ Pit[1,2] (Step 2)	Pit = [2,1] (uniquely identified)
4	At [1,2]: Stench=T, Breeze=F	Rule 1 (Stench→Wumpus adjacent); Rule 2 contrapositive ($\neg$ Breeze[1,2] $\Rightarrow$ no Pit adjacent to [1,2])	Wumpus $\in$ {[1,1],[1,3],[2,2]}, and since [1,1] already Safe $\Rightarrow$ Wumpus $\in$ {[1,3],[2,2]}; also reconfirms Pit $\notin$ {[1,1],[1,3],[2,2]}, consistent with Pit=[2,1]
5	Agent moves to [2,2] and survives; Glitter=T, Stench=F, Breeze=F	Occupied-cell convention + Rule 3	$\neg$ Wumpus[2,2] confirmed $\Rightarrow$ Safe[2,2]=True; Gold[2,2]=True
6	Combine Step 4 + Step 5	Elimination: Wumpus $\in$ {[1,3],[2,2]} and $\neg$ Wumpus[2,2] (Step 5)	Wumpus = [1,3] (uniquely identified)

Final located hazards

Pit is definitively at [2,1]. Wumpus is definitively at [1,3]. Gold is at [2,2]. All other visited cells — [1,1], [1,2], [2,2] — are confirmed Safe.

(c) Safety Evaluation of the Path [1,1] → [1,2] → [2,2]

Two levels of analysis are worth distinguishing, because they give different confidence levels and this distinction is often where correctness is lost in a rushed answer:

(i) A-priori view — before the agent has actually stepped anywhere

From [1,1] alone, Breeze[1,1] only tells the agent that the Pit is somewhere in {[1,2], [2,1]} — it does not, by itself, rule out [1,2]. A cautious agent applying pure logic at this point could not certify [1,2] as 100% safe; it is a candidate-risk cell (roughly a 50/50 possibility in the absence of further information, e.g. under the standard probabilistic-Wumpus-World tie-breaking heuristic).

(ii) A-posteriori view — using everything derived in (a) and (b)

Once the full percept sequence is taken into account (i.e., after Stench[1,2] and the fact that the agent survived at [1,2], and similarly for [2,2]), the elimination reasoning in part (b) removes all ambiguity:

- [1,2] is proven Safe (Step 2/3 above: Pit is uniquely [2,1], not [1,2]; and no Wumpus was ever a candidate there).
- [2,2] is proven Safe (Step 5/6 above: the agent occupies it unharmed and Wumpus is uniquely [1,3]).
- Neither hazard cell — Pit[2,1] nor Wumpus[1,3] — lies on the route [1,1]→[1,2]→[2,2].

Verdict

Given the complete derived knowledge state (parts a & b), the path [1,1] → [1,2] → [2,2] is SAFE, and the agent is logically justified in traversing it to collect the Gold at [2,2]. The caveat is procedural, not logical: a real agent must acquire the [1,2] and [2,2] percepts before it can certify them, so in a strict step-by-step plan it should treat [1,2] as an unconfirmed candidate-risk cell right up until the moment it senses no Stench/Breeze problem is disproven there — after which, as shown above, the danger is conclusively resolved in its favour.

Comparative Summary of Reasoning Techniques

Case Study	Logic Used	Primary Technique(s)	Verified Result
1. Medical Diagnosis	Propositional Logic	Forward Chaining, Backward Chaining	Flu = True, Pneumonia = True (both methods agree)
2. Traffic Management	First-Order Logic	Unification, Resolution Refutation	Proceed(CarA) proved True via empty clause □
3. Agricultural Expert System	Propositional Logic	CNF Conversion, Resolution Refutation, Non-monotonic reasoning	ApplyDripMethod proved True; CropAtRisk depends on monotonic vs. CWA convention once SoilDry is removed
4. Wumpus World	Propositional Logic (situated)	Modus Ponens, Forward Chaining, Elimination	Pit=[2,1], Wumpus=[1,3], Gold=[2,2]; path is safe

Conclusion

Across all four case studies, forward chaining, backward chaining, and resolution refutation consistently produce the same conclusions whenever they are applied to the same definite-clause knowledge base — which is exactly the guarantee that soundness and completeness of these inference procedures provide for Horn-clause logic. The exercises also surface two recurring themes that matter for real deployments of logical agents:

- Representational sufficiency is scenario-dependent. A KB that is complete for a single object (one patient, one emergency vehicle, one field) can silently become incomplete once multiple interacting objects are considered together, as shown by the two-ambulance analysis in Case Study 2 — additional predicates for conflict, priority, and time are required.
- The monotonic/non-monotonic distinction changes real answers. Case Study 3 shows that removing a single fact does not merely 'remove one conclusion' — whether a dependent conclusion (CropAtRisk) still holds depends entirely on whether the system uses classical monotonic entailment or a negation-as-failure (closed-world) convention, and expert-system designers must state this convention explicitly.
- Situated agents (Case Study 4) must combine positive and negative percepts. Safety in the Wumpus World was only fully established once the absence of a stench or breeze was also exploited (via the contrapositive of the rules), not just the presence of one — a purely positive reading of the rules leaves candidate hazard cells unresolved.

Taken together, these four case studies demonstrate the complete inference toolkit — propositional and first-order representation, forward and backward chaining, unification, and resolution refutation — applied correctly and cross-checked for consistency in every case.

Appendix A — Standard Inference Rules Used in This Document

For reference and to make every derivation step independently checkable, the standard inference rules invoked throughout the four case studies are summarised below.

Rule Name	Schema	Where Used
Modus Ponens	$p \rightarrow q, p \vdash q$	Case Studies 1, 2, 4 — the primary forward-reasoning step every time a rule's premise is confirmed true.
Modus Tollens (contrapositive)	$p \rightarrow q, \neg q \vdash \neg p$	Case Study 4(b) — absence of Stench/Breeze is used to rule out Wumpus/Pit in adjacent cells.
And-Elimination	$p \wedge q \vdash p \text{ (and)} \vdash q$	Case Study 1(c) — splitting a conjunctive rule premise into two independent sub-goals during backward chaining.
And-Introduction	$p, q \vdash p \wedge q$	Case Study 3(a)/(b) — combining SoilDry and CropWheat facts to satisfy conjunctive premises.
Unit Resolution	$p \vee q, \neg p \vdash q$	Case Studies 2(b) and 3(b) — the atomic step used repeatedly inside every full resolution derivation.
General Resolution	$(a1 \vee \dots \vee a_k \vee p), (b1 \vee \dots \vee b_m \vee \neg p) \vdash (a1 \vee \dots \vee a_k \vee b1 \vee \dots \vee b_m)$	Case Studies 2(b) and 3(b) — the general clause-vs-clause resolution rule; unification finds θ before this rule is applied in FOL.

Rule Name	Schema	Where Used
Elimination by Case Analysis	From a disjunctive candidate set {A,B} and a proof of $\neg A$, conclude B	Case Study 4(b) — narrowing $Pit \in \{[1,2],[2,1]\}$ down to the unique location [2,1], and similarly for the Wumpus.

Soundness and completeness. Modus Ponens, And-Elimination/Introduction and Resolution are all sound (they never derive a false conclusion from true premises). Forward chaining, backward chaining and resolution refutation are additionally complete for knowledge bases expressed as definite/Horn clauses (Case Studies 1, 3, and the relevant fragment of Case Study 4) and for general clausal form (Case Study 2) — meaning that if a sentence is truly entailed by the KB, these procedures are guaranteed to find a proof of it in finite time. This guarantee is exactly why every result in this document could be cross-checked by more than one method and found to agree.

Appendix B — Glossary of Key Terms

Term	Definition
Knowledge Base (KB)	The set of logical sentences (rules and facts) that a logical agent treats as known truths about its domain.
Proposition	An atomic statement that is either True or False, with no internal structure (e.g., $F = \text{'Patient has Fever'}$).
Predicate	In First-Order Logic, a relation over one or more objects that evaluates to True or False, e.g. $\text{Behind}(x, y)$.
Quantifier ($\forall, \exists$)	$\forall$ ('for all') asserts a sentence holds for every object in the domain; $\exists$ ('there exists') asserts it holds for at least one object.
Horn Clause / Definite Clause	A clause with at most one positive literal; rules of the form $\text{premise}_1 \wedge \text{premise}_2 \wedge \dots \rightarrow \text{conclusion}$ are definite clauses, the form used throughout Case Studies 1, 3 and 4.
Conjunctive Normal Form (CNF)	A sentence written as a conjunction of clauses, each clause being a disjunction of literals — the required input form for the resolution procedure.
Literal	An atomic proposition or its negation, e.g. SoilDry or $\neg \text{SoilDry}$.
Unification	The process of finding a substitution θ that makes two first-order expressions syntactically identical.
Most General Unifier (MGU)	Among all unifiers that make two expressions identical, the one that imposes the fewest possible constraints on the variables.
Forward Chaining	Data-driven inference: repeatedly fire any rule whose premises are already known, until no new facts are produced (fixpoint).
Backward Chaining	Goal-driven inference: start from a query and recursively reduce it to sub-goals via matching rule conclusions, until every branch is a known fact.

Term	Definition
Resolution Refutation	A proof-by-contradiction technique: negate the goal, add it to the CNF knowledge base, and resolve clauses until the empty clause ($\square$) is derived, proving the original goal was entailed.
Empty Clause ($\square$)	The result of resolving two complementary unit clauses (p and $\neg p$); its derivation signals a logical contradiction.
Closed-World Assumption (CWA)	A convention under which any sentence not provably true is assumed False — the basis of negation-as-failure reasoning in most rule-based expert systems.
Monotonicity	A property of classical logic: adding new facts to a KB can only add conclusions, never retract previously derived ones. CWA-based systems are non-monotonic.
Soundness	An inference procedure is sound if every conclusion it derives is guaranteed to be logically entailed by the premises.
Completeness	An inference procedure is complete if it is guaranteed to derive every sentence that is logically entailed by the premises.